

Topic 1: Introduction to Programming and C++

1.1 What is Programming?

- A computer program is a set of instructions given to a computer to perform a specific task
- A computer follows instructions step by step from top to bottom in sequence
- The programmer's role: writing, testing, and fixing those instructions
- Hardware: physical parts of a computer — CPU, RAM, keyboard, monitor
- Software: programs and data that run on hardware

1.2 Types of Programming Languages

- Machine language: binary code (0s and 1s) the CPU understands directly — not human-readable
- Assembly language: short codes like MOV, ADD that map directly to machine instructions
- High-level languages: English-like syntax — Python, Java, C# — easier to write and read
- Middle-level languages: combine high-level readability with low-level control — C and C++
- Compiled languages: entire source code translated to machine code before running — C, C++, Go
- Interpreted languages: translated and run line by line at runtime — Python, JavaScript
- Compiled programs are faster; interpreted programs are more flexible

1.3 Introduction to C++

- C++ was developed by Bjarne Stroustrup at Bell Labs starting in 1979
- Originally called "C with Classes", renamed to C++ in 1983
- The ++ comes from the increment operator — meaning an improved version of C
- C++ is a middle-level language: supports both high-level OOP and low-level memory control
- Used in: game engines (Unreal Engine), operating systems, embedded systems, competitive programming
- C++ is mostly backward compatible with C — valid C programs often compile as C++

- C++ adds: object-oriented programming, templates, exceptions, and the standard library
 - Major standards: C++98 (first standardized), C++11 (auto, nullptr, lambda), C++17 (further improvements)
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